

| Course Code                                                                                                                                                                                                                            | Course Title                           | L                | T | P | C |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------------------|---|---|---|
| BCSE410L                                                                                                                                                                                                                               | Cyber Security                         | 3                | 0 | 0 | 3 |
| Pre-requisite                                                                                                                                                                                                                          | NIL                                    | Syllabus version |   |   |   |
|                                                                                                                                                                                                                                        |                                        | 1.0              |   |   |   |
| Course Objectives                                                                                                                                                                                                                      |                                        |                  |   |   |   |
| 1. To understand the need for cybersecurity for solving the real word problems                                                                                                                                                         |                                        |                  |   |   |   |
| 2. To aware of ethical hacking methodologies for protecting cyber-physical systems.                                                                                                                                                    |                                        |                  |   |   |   |
| 3. To familiarize the defensive mechanisms, countermeasures, and best practices.                                                                                                                                                       |                                        |                  |   |   |   |
|                                                                                                                                                                                                                                        |                                        |                  |   |   |   |
| Course Outcomes                                                                                                                                                                                                                        |                                        |                  |   |   |   |
| 1. Understand the emerging cybersecurity attacks and their adversarial risk                                                                                                                                                            |                                        |                  |   |   |   |
| 2. Identify the emerging vulnerabilities and attacks, and countermeasures in cyber-physical systems.                                                                                                                                   |                                        |                  |   |   |   |
| 3. Comprehend the need for ethical hacking to minimize the security risk                                                                                                                                                               |                                        |                  |   |   |   |
| 4. Know the emerging security solutions using automated tools and techniques                                                                                                                                                           |                                        |                  |   |   |   |
|                                                                                                                                                                                                                                        |                                        |                  |   |   |   |
| Module:1                                                                                                                                                                                                                               | Foundation for Cyber Security          | 4 hours          |   |   |   |
| Hacker - Ethical hacker - Cyber-attacks: Network infrastructure attacks, Operating system attacks, Application and other specialized attacks - Security Assessment Principles                                                          |                                        |                  |   |   |   |
| Module:2                                                                                                                                                                                                                               | Hacking Methodology                    | 5 hours          |   |   |   |
| Methodology: Scanning the Systems and Network - Attack tree analysis - Assessing Vulnerabilities - Penetration Testing - Security Testing tools                                                                                        |                                        |                  |   |   |   |
| Module:3                                                                                                                                                                                                                               | Social Engineering                     | 7 hours          |   |   |   |
| Social Engineering Implications - Performing Social Engineering Attacks - Social Engineering Countermeasures: Policies, User awareness and training - Social Engineering Tool kit - Physical Security                                  |                                        |                  |   |   |   |
| Module:4                                                                                                                                                                                                                               | Password Security                      | 7 hours          |   |   |   |
| Password Vulnerabilities - Passwords Cracking Tools - Brute-force attacks - Rainbow attack - Password Cracking Countermeasures - Password Policy - Securing Operating Systems - Keyloggers tools                                       |                                        |                  |   |   |   |
| Module:5                                                                                                                                                                                                                               | Wireless and Mobile Security           | 7 hours          |   |   |   |
| Wireless and mobile Vulnerabilities and Attacks - Encrypted Traffic and countermeasures - Rogue wireless devices and countermeasures - MAC spoofing and countermeasures - Securing wireless workstations, Wi-Fi and Internet of Things |                                        |                  |   |   |   |
| Module:6                                                                                                                                                                                                                               | Operating System Security              | 6 hours          |   |   |   |
| OS Vulnerabilities: Windows, Linux and Mac - Detecting Null Sessions - Exploiting Missing Patches – Metasploit - Burp suite - Countermeasures against Buffer overflow and NFS attacks                                                  |                                        |                  |   |   |   |
| Module:7                                                                                                                                                                                                                               | Web Application and Databases Security | 7 hours          |   |   |   |
| Web App Security: Seeking out Web Vulnerabilities - Directory traversal - Input-filtering attacks - Code injection, SQL injection, Cross-site scripting Counter                                                                        |                                        |                  |   |   |   |

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| measures - Database Security: Database vulnerabilities - Minimizing Database Security Risks and Storage Security Risks – Counter measures and tools |                                                                                                                                                     |                             |                 |
| <b>Module:8</b>                                                                                                                                     |                                                                                                                                                     | <b>Recent Trends</b>        |                 |
|                                                                                                                                                     |                                                                                                                                                     | <b>2 hours</b>              |                 |
| Guest lectures from Industry and, Research and Development Organizations                                                                            |                                                                                                                                                     |                             |                 |
|                                                                                                                                                     |                                                                                                                                                     |                             |                 |
|                                                                                                                                                     |                                                                                                                                                     | <b>Total Lecture hours:</b> |                 |
|                                                                                                                                                     |                                                                                                                                                     | <b>45 hours</b>             |                 |
| <b>Text Book(s)</b>                                                                                                                                 |                                                                                                                                                     |                             |                 |
| 1.                                                                                                                                                  | Kevin Beaver CISSP, Hacking for Dummies, 2022, John Wiley & Sons, Inc, 7th Edition                                                                  |                             |                 |
| <b>Reference Books</b>                                                                                                                              |                                                                                                                                                     |                             |                 |
| 1.                                                                                                                                                  | Nina Godbole, SunitBelapure, Cyber Security, Understanding cybercrimes, computer forensics and legal perspectives, Reprint 2016, Wiley Publications |                             |                 |
| 2                                                                                                                                                   | Brooks, Charles J., Christopher Grow, Philip Craig, and Donald Short, Cybersecurity essentials, 2018, John Wiley & Sons,                            |                             |                 |
| 3.                                                                                                                                                  | Sammons, John, and Michael Cross. The basics of cyber safety: computer and mobile device safety made easy, 2016, Elsevier.                          |                             |                 |
| 4                                                                                                                                                   | Charles P. Pfleeger, Shari Lawrence, Pfleeger Jonathan Margulies; Security in Computing, 2015, Pearson Education Inc., 5th Edition.                 |                             |                 |
| Mode of Evaluation: CAT, Assignment, Quiz, FAT                                                                                                      |                                                                                                                                                     |                             |                 |
| Recommended by Board of Studies                                                                                                                     |                                                                                                                                                     | 12-05-2023                  |                 |
| Approved by Academic Council                                                                                                                        |                                                                                                                                                     | No. 70                      | Date 24-06-2023 |